22.3 Quantisation of Energy

Question Paper

Course	CIE A Level Physics
Section	22. Quantum Physics
Topic	22.3 Quantisation of Energy
Difficulty	Medium

Time allowed: 50

Score: /34

Percentage: /100

Question 1a

The energy levels of a hydrogen atom are shown in Fig. 1.1.

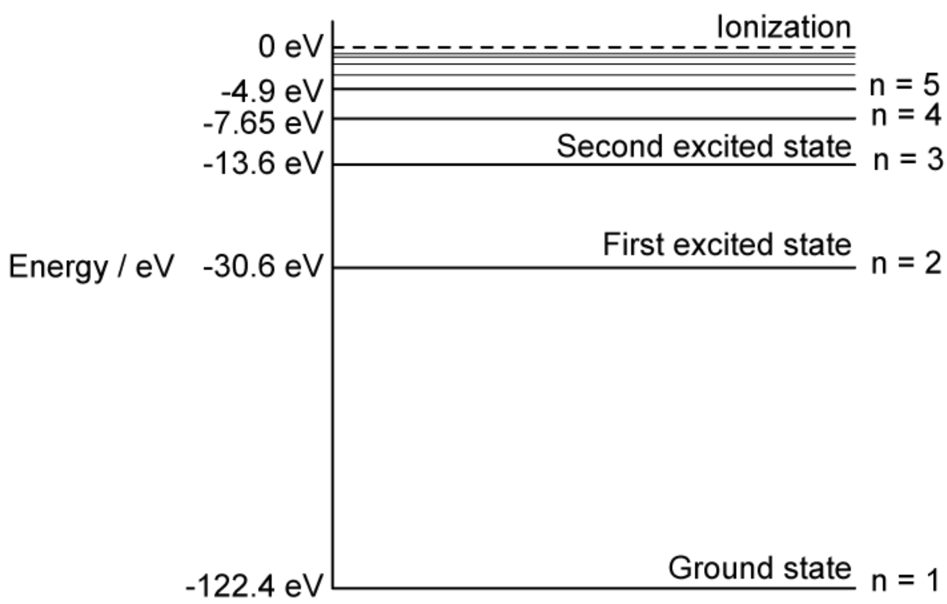


Fig. 1.1

Calculate the wavelength of radiation emitted when an electron falls from level *n* = 4 to the ground state in the hydrogen atom.

[4 marks]

Question 1b

When an electron of energy 1.84×10^{-17} J collides with a hydrogen atom, photons of six different energies are emitted.

Sketch arrows on Fig. 1.1 to show the transitions responsible for these photons.

[3 marks]

Question 1c

Calculate the wavelength of the photon with the smallest energy. Give your answer to an appropriate number of significant figures.

[3 marks]

Question 1d

One way to excite a hydrogen atom is by the absorption of a photon.

Explain why, for a particular transition, the photon must have an exact amount of energy.

[2 marks]

Question 2a

The lowest energy levels of a hydrogen atom are represented in Fig. 1.1.

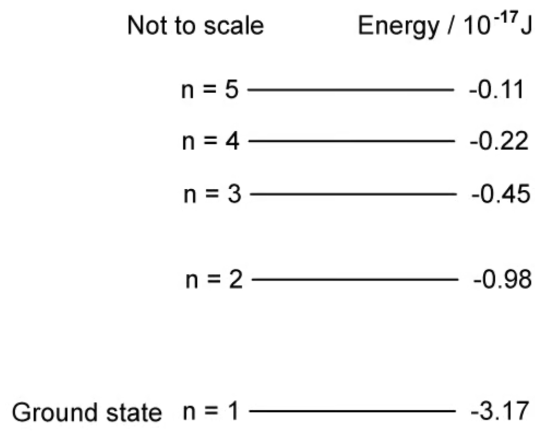


Fig 1.1

Identify the transition that produces a photon with frequency of 3.3×10^{16} Hz.

[2 marks]

Question 2b

An excited hydrogen atom can emit photons of certain discrete frequencies in various regions of the electromagnetic spectrum. Three possible transitions are shown in Fig. 1.2.

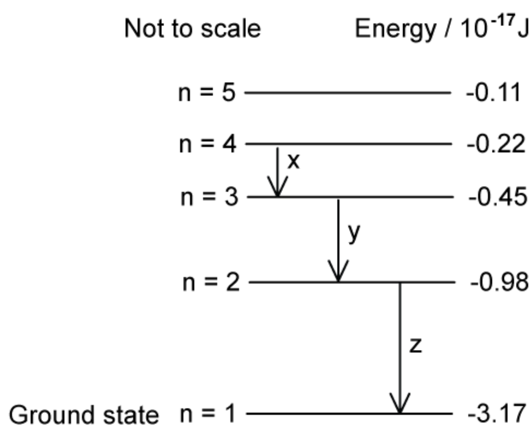


Fig. 1.2

Identify the transitions in Fig. 1.2 which produce a photon in the following regions of the electromagnetic spectrum:

(i)
Ultraviolet

[1]

(ii)
Visible

[1]

(iii)
Infrared

[1]

[3 marks]

Question 2c

Explain why the emitted photons in part **(b)** have certain discrete frequencies.

[3 marks]

Question 2d

An electron in the hydrogen atom is ionised by a photon of light with frequency 5.2×10^{16} Hz.

Calculate the kinetic energy of the ionised electron.

kinetic energy .... J

[3 marks]

Question 3a

Transitions between three energy levels in a particular atom give rise to three spectral lines.

In decreasing magnitudes, these are related to the wavelengths, λ_1, λ_2, and λ_3,

Use a diagram to show that the equation that relates λ_1, λ_2, and λ_3 is:

$$\frac{1}{\lambda_1} = \frac{1}{\lambda_2} + \frac{1}{\lambda_3}$$

[5 marks]

Question 3b

An atom has a ground state energy of –8.80 eV. The complete line emission spectra for this atom is shown in Fig. 1.1.

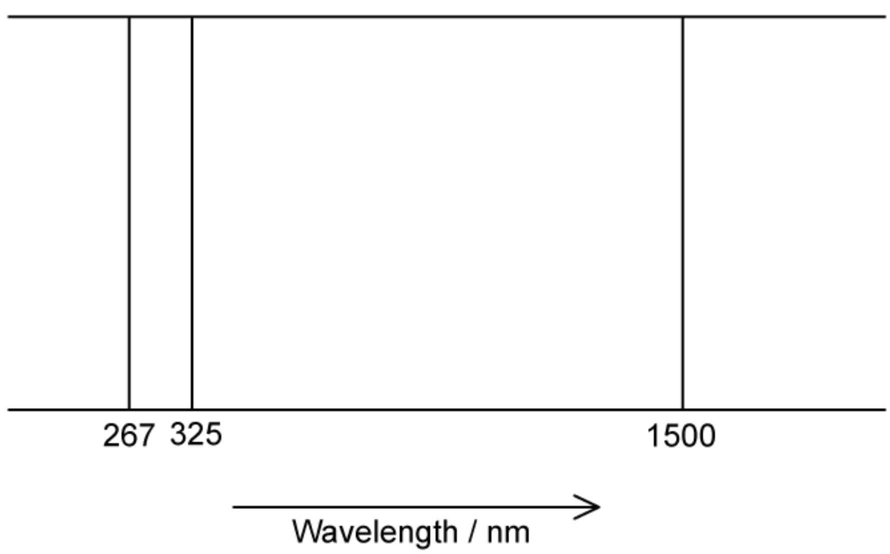


Fig. 1.1

Sketch a diagram of the possible energy levels for the atomic line spectra shown in Fig 1.1. Label each energy level with the value of its energy.

[6 marks]

Model Answers: Medium

1a

(a) Calculate the wavelength of radiation emitted when an electron falls from level n = 4 to the ground state in the hydrogen atom:

List the known quantities:

- Speed of light in free space, $c = 3.00 \times 10^8$ m s^{-1}
- Planck constant, $h = 6.63 \times 10^{-34}$ J s
- Electron energy at $n = 4$, $E_4 = -7.65$ eV
- Electron energy at $n = 1$, $E_1 = -122.4$ eV

Calculate the change in energy in J:

- $\Delta E = E_4 - E_1$
- $\Delta E = -7.65 - (-122.4) = 114.75$ eV
- $\Delta E = 114.75 \times (1.602 \times 10^{-19}) = 1.84 \times 10^{-17}$ J [1 mark]

Use the photoelectric equation to calculate the frequency of the electrons:

- $\Delta E = hf$
- $f = \frac{\Delta E}{h} = \frac{1.84 \times 10^{-17}}{6.63 \times 10^{-34}}$
- $f = 2.78 \times 10^{16}$ Hz [1 mark]

Use the wave equation to calculate the wavelength of the radiation emitted during the change of state:

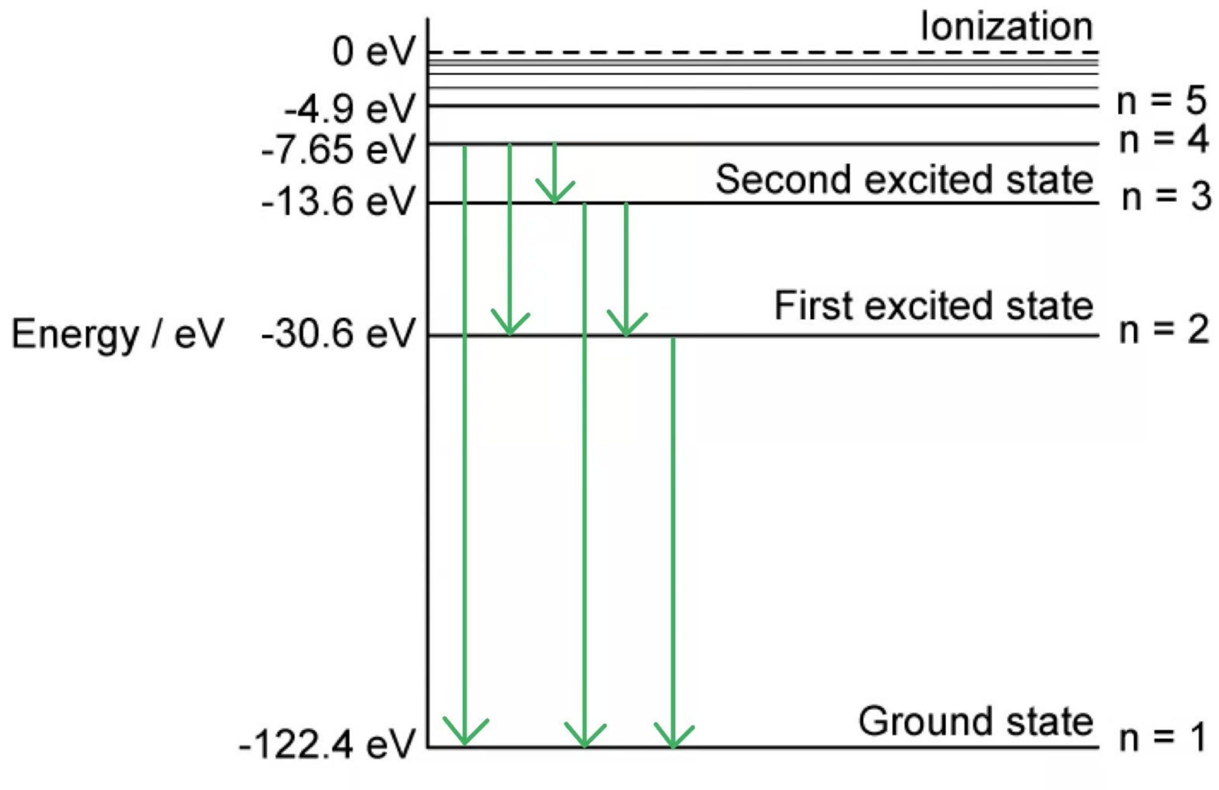
- $c = f\lambda$

- $\lambda = \frac{c}{f} = \frac{3 \times 10^8}{2.78 \times 10^{16}}$ [1 mark]
- $\lambda = 1.08 \times 10^{-8}$ m [1 mark]

[Total: 4 marks]

1b

(b) Arrows on the diagram that would score 4 marks to show the transitions responsible for the photons should look like this:



One mark awarded for showing **all** the transitions from each energy level:

- From $n = 4$:
  - $n = 4 \rightarrow n = 1$
  - $n = 4 \rightarrow n = 3$
  - $n = 4 \rightarrow n = 2$ [1 mark]
- From $n = 3$:
  - $n = 3 \rightarrow n = 2$

    - $n = 3 \rightarrow n = 1$ [1 mark]
- From $n = 2$:
    - $n = 2 \rightarrow n = 1$ [1 mark]

[Total: 3 marks]

Remember, the **total energy** of the transition from $n = 4$ to $n = 1$ equals the **sum** of all the transitions in between.

1c

(c) Calculate the wavelength of the photon with the smallest energy:

List the known quantities:

- Speed of light in free space, $c = 3.00 \times 10^8$ m s^{-1}
- Planck constant, $h = 6.63 \times 10^{-34}$ J
- Energy at $n = 4$, $E_4 = -7.65$ eV
- Energy at $n = 3$, $E_3 = -13.6$ eV

Calculate the smallest change in energy in J from $n = 4$ to $n = 3$:

- $\Delta E = E_4 - E_3$
- $\Delta E = (-7.65) - (-13.6) = 5.95 \text{ eV}$
- $\Delta E = 5.95 \times (1.602 \times 10^{-19}) = 9.53 \times 10^{-19} \text{ J}$ [1 mark]

Use the photoelectric equation to calculate the wavelength of the photon:

- $\Delta E = hf = \dfrac{hc}{\lambda}$
- $\lambda = \dfrac{hc}{\Delta E} = \dfrac{(6.63 \times 10^{-34}) \times (3.00 \times 10^8)}{(9.53 \times 10^{-19})}$ [1 mark]
- $\lambda = 2.087 \times 10^{-7} \text{ m} = 2.09 \times 10^{-7} \text{ m} \, (3 \text{ s.f.})$ [1 mark]

[Total: 3 marks]

1d

(d) The photon must have an exact amount of energy for a particular transition because:

Any **two** from:

- Energy is needed for an electron to move to a higher level; [1 mark]
- For a transition / excitation / change of energy levels, an exact amount of energy is required; [1 mark]
- The photon's energy must be equal to the difference between the energy levels; [1 mark]
- All of the photon's energy is absorbed (in a on-to-one interaction); [1 mark]

[Total: 2 marks]

2a

(a) Identify the transition that produces a photon with frequency of 3.3×10^{16} Hz:

List the known quantities:

- Frequency of photon, $f = 3.3 \times 10^{16}$ Hz
- Planck constant, $h = 6.63 \times 10^{-34}$ J s

Use the photoelectric equation to calculate the energy change:

- $\Delta E = hf$
- $\Delta E = \left(6.63 \times 10^{-34}\right) \times \left(3.3 \times 10^{16}\right)$
- $\Delta E = 2.19 \times 10^{-17}$ J [1 mark]

Consider the transitions between the energy states:

- $\Delta E = \left(-0.98 \times 10^{-17}\right) - \left(-3.17 \times 10^{-17}\right) = 2.19 \times 10^{-17}$ J
- $\Delta E = E_2 - E_1$ [1 mark]

OR

- So, the transition is from $n = 2$ to the ground state; [1 mark]

[Total: 2 marks]

OR

- So, the transition is from $n = 2$ to the ground state; [1 mark]

[Total: 2 marks]

Use **trial and error** through all the combinations to find the correct **energy transition**.

2b

(b) Identify the transitions in Fig. 1.2 which produce a photon in the following regions of the electromagnetic spectrum:

Consider the smallest energy change in transition X:

- $\Delta E_X = (-0.22 \times 10^{-17}) - (-0.45 \times 10^{-17}) = 0.23 \times 10^{-17}$ J
- Lower energy = lower frequency and a longer wavelength
- Infrared radiation has the longest wavelength

Consider the largest energy change in transition Z:

- $\Delta E_Z = (-0.98 \times 10^{-17}) - (-3.17 \times 10^{-17}) = 2.19 \times 10^{-17}$ J
- Higher energy = higher frequency and a shorter wavelength
- Ultraviolet radiation has the shortest wavelength

Use the process of elimination to identify the energy change in Y:

- This must be visible light

(i) Ultraviolet radiation is represented by:

- Transition Z; [1 mark]

(ii) Visible light is represented by:

- Transition Y; [1 mark]

(iii) Infrared radiation is represented by:

- Transition X; [1 mark]

[Total: 3 marks]

2c

(c) The emitted photons in part **(b)** have certain discrete frequencies because:

- (Electromagnetic radiation is emitted when) an electron moves from one fixed energy level to a lower fixed energy level; [1 mark]
- The (emitted) photon is given a discrete / specific amount of energy; [1 mark]
- The photon energy is directly proportional to the frequency / $\Delta E = hf$, therefore the emitted photons have discrete frequencies; [1 mark]

[Total: 3 marks]

2d

(d) Calculate the kinetic energy of the ionised electron:

List the known quantities:

- Frequency of photon, $f = 5.2 \times 10^{16}$ Hz
- Planck constant, $h = 6.63 \times 10^{-34}$ J s

Use Fig. 1.1 in **(a)** to identify the ionisation energy of the hydrogen atom:

- $E_i = 3.17 \times 10^{-17}$ J [1 mark]

Use the photoelectric equation to calculate the energy absorbed from the photon:

- $E_p = hf$
- $E_p = (6.63 \times 10^{-34}) \times (5.2 \times 10^{16})$
- $E_p = 3.44 \times 10^{-17}$ J [1 mark]

Calculate the kinetic energy of the electron:

- $KE = E_p - E_i$
- $KE = (3.44 \times 10^{-17}) - (3.17 \times 10^{-17})$
- $KE = 2.7 \times 10^{-18}$ J [1 mark]

[Total: 3 marks]

Remember that the **energy** in this interaction is **conserved**.

Energy before absorption = Energy after absorption

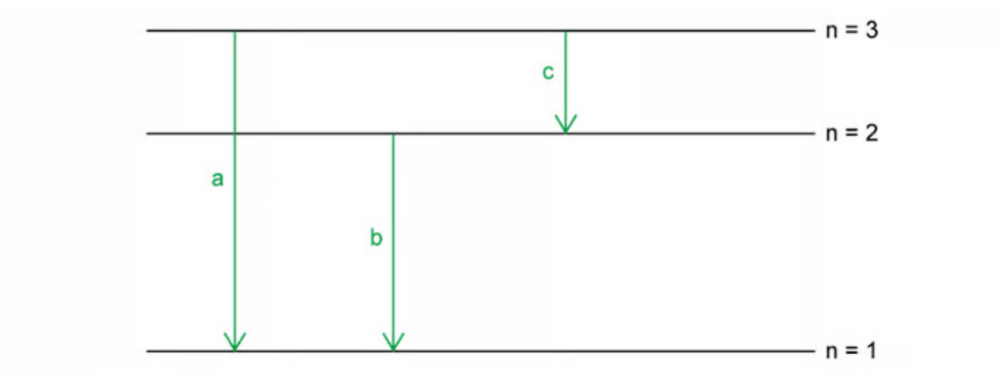
Energy absorbed from photon = electron Ionisation energy + KE of electron

The electron ionisation energy = energy required to change energy level

KE of electron is the energy the electron moves with

3a

(a) A diagram to show that the equation that relates λ_1, λ_2, and λ_3 that would score 2 marks should look like this:



- Three energy levels shown with a smaller gap size between the first two levels; [1 mark]
- The three energy levels are labelled according to the three photons of decreasing energy / wavelength; [1 mark]

Consider the energy level diagram and how it relates to spectral lines of different wavelengths:

- The energy level diagram shows three energy transitions that result in photons of different energy (and hence wavelength)
- The energy of transition (a) is the largest and the energy of transition (c) is the smallest
- Therefore (a) corresponds to spectral line of wavelength λ_1; [1 mark]

- (b) corresponds to a spectral line of wavelength λ_2 **AND** (c) corresponds to a spectral line of wavelength λ_3; [1 mark]

Use the photoelectric equation to link the photon energies and the spectral lines:

- From the energy level diagram: Energy released in transition (a) = energy released in transition (b) + energy released in transition (c)
- This can be written as: $E_1 = E_2 + E_3 = \frac{hc}{\lambda_1} = \frac{hc}{\lambda_2} + \frac{hc}{\lambda_3}$ and therefore

  $\frac{1}{\lambda_1} = \frac{1}{\lambda_2} + \frac{1}{\lambda_3}$; [1 mark]

[Total: 5 marks]

The **final step** in the rearrangement of the equation is:

$$\frac{hc}{\lambda_1} = \frac{hc}{\lambda_2} + \frac{hc}{\lambda_3}$$

$$\times\frac{1}{hc} \quad \boxed{\frac{1}{\lambda_1} = \frac{1}{\lambda_2} + \frac{1}{\lambda_3}} \quad \times\frac{1}{hc}$$

3b

(b) Sketch a diagram of the possible energy levels for the atomic line spectra shown in Fig 1.1:

List the known quantities:

- Ground state energy of the atom = –8.80 eV
- Speed of light in free space, $c = 3.00 \times 10^8$ m s^{-1}
- Planck constant, $h = 6.63 \times 10^{-34}$ J s
- Shortest wavelength photon emitted, $\lambda_a = 267$ nm $= 267 \times 10^{-9}$ m
- Middle wavelength photon emitted, $\lambda_b = 325$ nm $= 325 \times 10^{-9}$ m
- Longest wavelength photon emitted, $\lambda_c = 15\,00$ nm $= 1500 \times 10^{-9}$ m

Use the photoelectric equation to calculate the energy of λ_c:

[Total: 5 marks]

The **final step** in the rearrangement of the equation is:

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$$\times \frac{1}{hc} \qquad \boxed{\frac{1}{\lambda_1} = \frac{1}{\lambda_2} + \frac{1}{\lambda_3}} \qquad \times \frac{1}{hc}$$

3b

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- Planck constant, $h = 6.63 \times 10^{-34}$ J s
- Shortest wavelength photon emitted, $\lambda_a = 267$ nm = 267×10^{-9} m
- Middle wavelength photon emitted, $\lambda_b = 325$ nm = 325×10^{-9} m
- Longest wavelength photon emitted, $\lambda_c = 15\,00$ nm = 1500×10^{-9} m

Use the photoelectric equation to calculate the energy of λ_c:

- $E = hf = \dfrac{hc}{\lambda}$
- $E_c = \dfrac{(6.63 \times 10^{-34}) \times (3 \times 10^8)}{1500 \times 10^{-9}} = 1.33 \times 10^{-19}$ J
- In eV, this is $\dfrac{1.33 \times 10^{-19}}{1.6 \times 10^{-19}} = 0.83$ eV [1 mark]

Calculate the first energy level, n = 2:

- Ground state energy level n = 1, E = – 8.8 eV

- First energy level = ground state + energy emitted in transition for λ_c
- First energy level = – 8.8 + 0.83 = – 7.97 eV [1 mark]

Calculate the energy of λ_b:

- $E_b = \frac{(6.63 \times 10^{-34}) \times (3 \times 10^8)}{325 \times 10^{-9}} = 6.12 \times 10^{-19}$ J
- In eV, this is $\frac{6.12 \times 10^{-19}}{1.6 \times 10^{-19}} = 3.83$ eV [1 mark]

Calculate the second energy level, n = 3:

- Second energy level = first energy level + energy emitted in transition for λ_b
- Second energy level = – 7.97 + 3.83 = –4.14 eV [1 mark]

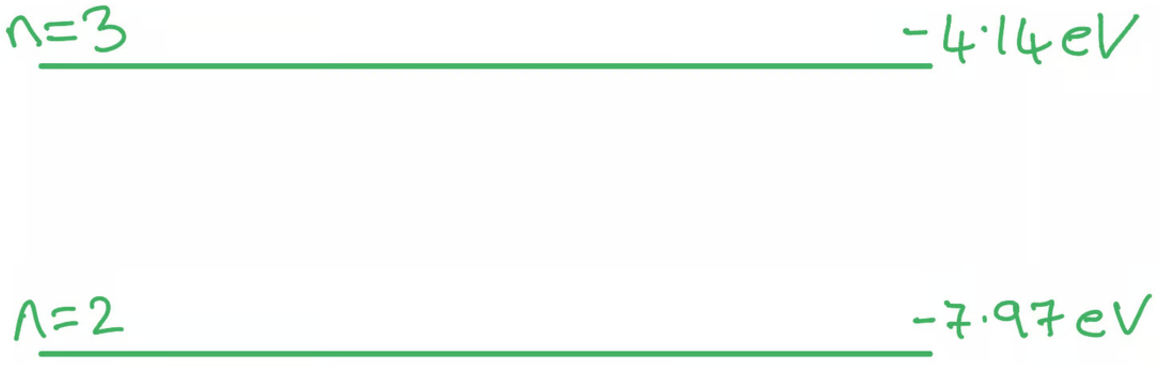
Calculate the energy of λ_a:

- $E_a = \frac{(6.63 \times 10^{-34}) \times (3 \times 10^8)}{267 \times 10^{-9}} = 7.46 \times 10^{-19}$ J
- In eV, this is $\frac{7.46 \times 10^{-19}}{1.6 \times 10^{-19}} = 4.66$ eV [1 mark]

Check your calculations at the end by calculating the value of the transition from n = 1 to n = 3:

- Ground state energy + energy of λ_a = energy level at n = 3 = –8.8 + 4.66 = –4.14 eV
- This is equal to the two transition energies $\lambda_b + \lambda_c$ = 3.83 + 0.83 = 4.66 eV

A diagram of the possible energy levels that would score 1 mark should look like this:


n=3
-4·14eV
n=2
-7·97eV